1. The following reaction

$$2H_{2(g)} + 2NO_{(g)} \rightarrow N_{2(g)} + 2H_2O_{(g)}$$

was found to be first order with respect to H_2 and second order with respect to NO

a) One possible mechanism was proposed to be

$$H_{2(g)} + 2NO_{(g)} \xrightarrow{k_1} N_2O_{(g)} + H_2O_{(g)}$$

$$H_{2(g)} + N_2O_{(g)} \xrightarrow{k_2} N_{2(g)} + H_2O_{(g)}$$

Under what conditions does this mechanism give the observed rate law? Express the observed rate constant, k_{obs}, in terms of the elementary rate constants

b) A second possible mechanism was proposed to be

$$2NO_{(g)} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} N_2O_{2(g)}$$

$$H_{2(g)} + N_2O_{2(g)} \xrightarrow{k_2} N_2O_{(g)} + H_2O_{(g)}$$

$$H_{2(g)} + N_2O_{(g)} \xrightarrow{k_3} N_{2(g)} + H_2O_{(g)}$$

Under what conditions does this mechanism give the observed rate law? Express k_{obs} in terms of the elementary rate constants?

c) Which of these two mechanisms do you think is more likely to be true? Explain your reasoning

d) Using your answer from (c), sketch a possible energy-vs-reaction coordinate diagram that is compatible with the mechansism.

e) Describe an experimental approach that could allow you to measure the rate of formation of the product(s).

[Question 2 on next page]

2. A chain reaction occurs when a reactive intermediate (often a radical) is generated by an initiation step, is used up in propagation step 1 and is then regenerated in propagation step 2.

The reaction

$$CH_3CHO_{(g)} \rightarrow CH_{4(g)} + CO_{(g)}$$

Is proposed to proceed via the following chain reaction (pay attention to which species are radicals)

Initiation: $CH_3CHO \xrightarrow{k_1} CH_3\cdot + CH_3CO\cdot$

Propagation step 1: $\cdot CH_3 + CH_3CHO \xrightarrow{k_2} CH_4 + CH_3CO\cdot$

Propagation step 2: $CH_3CO\cdot \xrightarrow{k_3} \cdot CH_3 + CO$

Termination: $\cdot CH_3 + \cdot CH_3 \xrightarrow{k_4} C_2H_6$

applying the steady state condition to the intermediates, express the rate law for the formation of CH_4